

NEURAL NETWORK ZOO

Neural networks are computer systems inspired by the human brain, designed to recognize patterns and learn from data. They are a major part of Deep Learning, a branch of Artificial Intelligence focused on training large models to make predictions, classify information, or generate outputs such as text or images. A neural network is made up of neurons (also called nodes) that are organized into layers:

- ➔ Input Layer - receives the raw data (images, numbers, text)
- ➔ Hidden Layers - process the data using mathematical weights and activation functions
- ➔ Output Layer - produces the final prediction or decision

Each neuron takes inputs, multiplies them by weights, adds a bias, and passes the result through an activation function to decide whether it should “fire.” When many neurons are connected in layers, they form a network capable of learning complex patterns.

The Neural Network Zoo is a creative learning concept where each type of neural network is represented as an animal. The idea is to make technical topics more memorable and engaging by associating each network with animal traits that match its strength.

Neural Network Type	Animal	Why This Animal
Convolutional Neural Network (CNN)	Cheetah	Fast and sharp vision, great at image recognition
Recurrent Neural Network (RNN)	Raccoon	Curious and good with sequences, remembers recent steps
Long Short-Term Memory	Lemur	Strong memory and balance, remembers long sequences
Transformer Network	Elephant	Massive memory and attention power
Autoencoder	Chameleon	Adapts and compresses information

CHOSEN NEURAL NETWORK: LONG SHORT-TERM MEMORY (LSTM)



Reflective Analysis: The Neural Network Zoo combines learning and creativity, allowing me to understand deep learning concepts in a memorable and engaging way, rather than just memorizing words and definitions. Despite each neural network having a different animal identity and unique strength, they all share the same basic foundation of layers, neurons, and learning from data. Seeing how CNNs focused on images, RNNs handled short sequences, and LSTMs managed long-term memory helped me understand that they are not one universal tool but a collection of specialized models designed for different types of problems. The metaphors definitely helped remember their purpose. Although I focused primarily on the LSTM, I also tried generating a poster for each animal.

